Operations Management Case Pack

Supply Chains on the Move: China → Malaysia, and the Geely × Proton Case

产业链从中国转移到马来西亚 · 吉利 × 宝腾

Student handout · Information as of October 2026

Contents
Part 1. The China → Malaysia supply-chain shift (overview)
Part 2. Case: Geely × Proton, 2017–2026
Part 3. Pre-reading questions by session
Part 4. Exercises A–D (with answer space)
Part 5. Class debate
Part 6. Sources and data notes

Figures marked † are approximate and drawn from public reporting. They were not re-verified against primary filings. All exercise numbers are illustrative teaching data, not company data.

Part 1. The China → Malaysia supply-chain shift

Over the past decade Malaysia has moved from absorbing low-margin "China+1" overflow, such as solar modules and electronics assembly, to hosting **strategic Chinese capacity**. That now includes lithium-battery cells and materials, right-hand-drive EV assembly, and one of the largest clusters of Chinese-operated data-centre capacity outside China.

1.1 Three waves

Wave	Period	Main driver	What moved
1. Cost and Belt & Road overflow	2013–2017	Chinese overcapacity, BRI financing, state partners	Steel (Kuantan industrial park), solar, industrial parks
2. Tariff evasion → hedging	2018–2022	US Section 301 tariffs, COVID shocks	Final assembly for US-bound goods
3. Strategic localisation	2023–2026	US/EU de-risking, chip export controls, Malaysia NIMP 2030 and NETR	Cell and materials plants, EV/CKD plants, hyperscale data centres

1.2 Key data

Indicator	Value
Malaysia approved investment, 2024	RM378.5bn (record; later revised to ~RM384bn)
Malaysia approved investment, 2025	RM426.7bn (record); foreign RM207.1bn (+20.9%)
China rank among foreign investors, 1H2025	#2 (RM23.4bn), behind Singapore
Chinese manufacturing approvals in Penang, 2025	RM4.2bn; more than 80 Chinese firms operating
Johor data-centre capacity, mid-2025	~580 MW live; ~5.8 GW in the pipeline
US reciprocal tariff on Malaysia (from Aug 2025)	19%; up to 40% for goods found to be transhipped
US AD/CVD on Malaysian solar cells (Apr 2025)	Up to 168.8%, depending on the company

1.3 Four sectors at a glance

- **New-energy batteries.** Chinese cell makers cluster in Kedah (Kulim, Gurun) and Johor. EVE Energy's Kulim plant started production in 2025, and EVE announced a ~US$1.2bn energy-storage plant. Materials suppliers followed: anodes (Putailai, US$297m), cathodes and electrolytes. Output is mostly cylindrical cells and energy-storage batteries, not EV traction cells.
- **Automotive.** Geely bought 49.9% of Proton in 2017 (see Part 2). BYD, Chery, GWM and others entered through imports and then local CKD assembly. BYD's own plant in Tanjung Malim is due in 2H2026.
- **Specialty chemicals.** This is the slowest-moving sector. It has a Petronas-anchored feedstock base, a large Chinese refining pledge (Rongsheng, Pengerang; status unconfirmed), and smaller additive and battery-chemical plants.

- **Data centres.** Singapore's 2019 pause on new data centres pushed demand into Johor. Chinese operators (GDS/DayOne, ByteDance-linked capacity, Alibaba, Huawei) moved early. Since 2025 the sector has faced AI-chip export permits, higher power tariffs and water limits.

1.4 Why Malaysia? Pull factors cited

- Geopolitical neutrality between the US and China
- A 50-year electronics and semiconductor base in Penang
- Ethnic-Chinese business networks and Mandarin-speaking managers
- Industrial land, power and water (historically cheap)
- Industrial policy: NIMP 2030, NETR, the National Semiconductor Strategy, time-limited duty exemptions on imported EVs
- Market access: RCEP and right-hand-drive export markets

Part 2. Case: Geely × Proton (2017–2026)

2.1 Before the deal

Proton was founded in 1983 as Malaysia's national car company using Mitsubishi technology. Its domestic share fell from more than 70% in its early years to about 12% in 2016. Problems included ageing platforms, a quality reputation gap against Japanese and Perodua models, a fragmented local supplier base, failed export attempts, and the capital needs of Lotus (bought in 1996).

Year	Proton sales / output (units)
2014	115,783
2015	102,175
2016	~72,290 (production; down ~30%)

In 2016 the government extended a **RM1.5bn soft loan** on condition that Proton found a strategic foreign partner. The candidates reportedly included PSA, Renault, Suzuki and Geely. Geely withdrew a bid for majority control in March 2017 and returned in May with a minority offer.

2.2 The deal (May–June 2017)

Element	Terms
Equity	Geely takes 49.9% of Proton; DRB-HICOM keeps 50.1%
Cash	About RM170.3m Geely investment
Technology	Boyue SUV platform transferred (valued ~RM290m); later platforms licensed
Restructuring	~RM1.2bn of non-automotive assets moved to DRB-HICOM; Proton repays RM567m to DRB-HICOM
Lotus	Geely 51%; Etika Automotive 49%

2.3 What changed after 2017

- **Products.** Geely-based models were added: X70 (Boyue, Dec 2018 †), X50 (2020 †), X90 and S70 (2023 †), e.MAS 7 (late 2024) and e.MAS 5 (2026). Proton's own Saga, Persona and Iriz stayed in the range.
- **Manufacturing and quality.** Geely manufacturing and quality systems were brought into Tanjung Malim. Geely executives held key operating roles at first (CEO Dr Li Chunrong, 2018–2021 †), followed by a Malaysian CEO.
- **Localisation path.** CBU import → CKD assembly → rising local content. The Tanjung Malim engine plant is being prepared for hybrid, PHEV and EV components (target 400,000 units by 2028).
- **Purchasing and suppliers.** Supplier base consolidated; Geely supplier-quality engineers worked with local vendors; parts co-sourced with Geely Holding's volume (more than 4 million vehicles in 2025).
- **Distribution and brand.** Dealer network upgraded; separate e.MAS EV brand; Proton International Corporation set up in July 2025 to run exports.

2.4 Results

Year	Proton sales (units)	Notes
2018	~65,000 †	X70 launched in December
2019	~100,000 †	
2020	~109,000 †	X50 launched; COVID year
2022	141,432	Highest since 2013; DRB-HICOM back to pre-tax profit
2023	~154,000 †	
2025	157,976	19.4% domestic share; exports ~6,000 (Egypt 2,280, Vietnam 1,316)
2026 (Jan–Sep)	160,886	~26–27% YTD share; target 200,000 for the full year

Market context in 2026: total industry volume is about 815,000 units a year. Japanese brands' share fell from ~35% to ~21%. Chinese brands (BYD, Chery/Jaecoo, Jetour, Zeekr and others) reached about 7.8% combined. EV penetration rose to ~9.2%.

2.5 Automotive High-Tech Valley (AHTV), Tanjung Malim

AHTV is a DRB-HICOM–Geely development worth about **RM32bn** over roughly 10 years. Geely's stated target is **500,000 vehicles a year by 2035, about half for export**, positioning Malaysia as its right-hand-drive hub. Proton's EV plant opened in September 2025 (20,000 units a year in phase 1, rising to 45,000). Zeekr assembly has been announced. BYD is building a separate plant nearby.

2.6 Open issues

- Proton's current competitive range is largely Geely-derived.
- EV battery cells, motors and power electronics are still mainly imported from China.
- Geely's own brands are also sold in Malaysia.
- Exports of Chinese-platform vehicles may face origin and security scrutiny in some markets.
- Proton's standalone profitability is not fully disclosed.

Part 3. Pre-reading questions

Session 1: Why Malaysia, and why now?

1. If China is cheaper and more capable, why are Chinese firms building factories in Malaysia?
2. Rank the pull factors in §1.4 for (a) a battery-cell maker, (b) an EV OEM, (c) a hyperscale data-centre operator. Why do the rankings differ?
3. Do the four sectors move for the same reasons? Which sector moved least, and why?

Session 2: Tariffs, origin and the solar precedent

1. What does "substantial transformation" mean, and why does it matter for a plant that imports most of its inputs?
2. Chinese solar makers relocated to Southeast Asia after 2012 and were hit by new US duties in 2022–2025. Is a battery plant in Kedah more or less exposed? Why?
3. A European OEM is considering a Chinese-owned battery supplier in Malaysia. What should it check beyond a standard supplier audit?

Session 3: Geely × Proton

1. In 2017 you are Proton's COO. Geely offers platforms, not cash. Do you accept? On what conditions?
2. List the operational levers Geely pulled (product, manufacturing, purchasing, distribution). Which would be hardest for a competitor to copy?
3. Why does Proton keep its own Saga instead of moving fully to Geely platforms?
4. Would the same approach work for BYD, which is building a wholly owned plant? What does each entry mode gain and lose?
5. What should the 2017 agreement have said about local R&D and engineering? Draft one clause.

Part 4. Exercises

All numbers below are illustrative teaching data.

Exercise A: Factor-rating location model

Score each country from 1 to 10 for a new battery-cell plant serving US and EU customers. The tariff row is pre-filled with illustrative scores; you may re-score it from current US tariff rates.

Factor	Weight	Malaysia	Vietnam	Thailand	Indonesia
Labour cost	0.10	6	8	6	8
Skilled technicians	0.15	8	6	7	5
Power cost and reliability	0.15	7	5	7	6
Tariff and origin exposure to the US	0.20	6	4	5	6
Supplier ecosystem	0.15	7	6	7	6
Political neutrality	0.10	8	6	7	7
Raw materials (nickel)	0.15	3	3	3	10
Weighted score	1.00				

1. Compute the weighted score for each country. Which ranks first?
2. Sensitivity: how much must the tariff weight rise (taking weight from other factors) before the ranking changes? Show your working.
3. What does the model leave out? Consider ownership-based rules (e.g. US "foreign entity of concern" tests) and battery chemistry (NMC vs LFP).

Working / answer:

Exercise B: Total landed cost under tariff scenarios

A battery module made in Malaysia has an ex-works cost of US$100 and freight to the US of US$6. 70% of its inputs are Chinese-origin. State your assumption about which value the duty applies to.

Scenario	Ex-works (US$)	Freight (US$)	Duty (US$)	Landed cost (US$)
1. Reciprocal tariff 19%	100	6		
2. Transhipment finding 40%	100	6		
3. AD/CVD order 100% (state whether it stacks on the 19%)	100	6		
4. Localise to 50% local inputs (+US$8 cost), origin secured, 19%	108	6		

1. Complete the table.
2. Let p be the probability of scenario 2. For what value of p does localisation (scenario 4) pay off? Repeat for scenario 3.
3. What non-cost benefits and risks of localising should the decision also weigh?

Working / answer:

Exercise C: The learning curve and the CBU → CKD switch

The first CKD vehicle at Tanjung Malim takes 100 labour-hours. Assume a 90% learning curve (unit-time model: $T_n = T_1 \cdot n^b$, where b = log 0.9 / log 2).

Unit n	Labour-hours T_n
1	100
1,000	
10,000	
Unit at which T_n = 30 hours	

1. Complete the table.
2. At a phase-1 rate of 20,000 vehicles a year, how many months does it take to reach 30 hours per vehicle?
3. Is labour learning or supplier localisation the real constraint on the ramp-up? How should this shape the timing of the CBU → CKD switch before the duty exemption ends?

Working / answer:

Exercise D: AHTV capacity break-even

The plant has fixed costs of RM1.8bn a year at 500,000 capacity and a contribution margin of RM6,000 per vehicle. The domestic Geely-group share is assumed to be 25% of an 815,000-unit market.

Item	25% domestic share	18% domestic share
Break-even volume (units)		
Utilisation at break-even		
Domestic volume (units)		
Export volume needed (units)		
Multiple of 2025 exports (~6,000)		

1. Complete the table.
2. Which assumptions in this model would you challenge first? What would make the 500,000 target credible?

Working / answer:

Part 5. Class debate

"Proton in 2026 is a Malaysian badge on a Chinese car."

The class splits into two teams, **for** and **against**. Each team has 5 minutes to open, 3 minutes to rebut, and 2 minutes to close. Use operational evidence: product platforms, bill-of-materials origin, manufacturing and quality systems, supplier base, management, and exports. Avoid national loyalties.

Preparation notes:

Working / answer:

Optional group project ideas

- Using UN Comtrade/BACI data, test whether Malaysian battery or solar exports to the US track rising imports of Chinese intermediates.
- Compare entry modes: Geely–Proton (minority stake), BYD (wholly owned), Chery (contract assembly), Great Wall (bought a plant in Thailand).
- Build a supply-chain diversification index that counts ownership and input origin, not just ship-from country.

Part 6. Sources and data notes

Data notes. Figures marked † are approximate values from public reporting and were not re-verified against primary filings. Data-centre capacity figures mix live, under-construction and planned capacity, depending on the source. All exercise data are illustrative. Check current figures against primary sources (MIDA, Malaysian Automotive Association, DRB-HICOM and Geely filings) before citing.

1. MIDA – blockbuster year for foreign investment. https://www.mida.gov.my/mida-news/stories-of-the-year-a-blockbuster-year-for-foreign-investment/
2. MIDA/MITI – RM17.9bn Chinese investments 2023–1H2024. https://www.mida.gov.my/mida-news/rm17-9b-investments-from-china-approved-from-2023-until-1h2024-miti/
3. MIDA – Chinese companies keen on Penang. https://www.mida.gov.my/mida-news/more-chinese-companies-keen-on-investing-in-penang-says-cm/
4. CnEVPost – EVE Energy Malaysia ESS project. https://cnevpost.com/2025/06/30/eve-energy-plans-energy-storage-battery-project-malaysia/
5. Yicai – Putailai anode plant. https://english.sse.com.cn/news/newsrelease/voice/c/c_20260312_10811607.shtml
6. JLL – Johor data centres Q2 2025. https://research.jllapsites.com/appd-market-report/q2-2025-data-centres-johor-bahru/
7. SCMP – Malaysia AI chip trade permits. https://www.scmp.com/week-asia/economics/article/3318099/malaysia-imposes-trade-permits-us-linked-ai-chip-shipments
8. Malay Mail – MITI tightens origin checks. https://www.malaymail.com/news/malaysia/2025/08/27/miti-tightens-origin-checks-as-us-delays-clarity-on-transhipment-enforcement-under-new-tariff-order/189013
9. NST – US solar tariffs on Malaysia. https://www.nst.com.my/amp/business/corporate/2025/04/1205626/malaysia-feel-pinch-new-us-solar-tariffs
10. Carlist – Geely acquires 49.9% of Proton. https://www.carlist.my/news/confirmed-geely-acquires-499-percent-stake-in-proton-plus-51-percent-of-lotus/62621/amp/
11. Just Auto – Geely agreement for Proton and Lotus. https://www.just-auto.com/news/geely-strikes-agreement-to-acquire-proton-and-lotus
12. FMT – RM1.5bn loan for Proton. https://www.freemalaysiatoday.com/category/nation/2016/03/18/poser-over-rm1-5-billion-loan-for-proton
13. FMT – Uphill climb for Proton despite Geely. https://fmtv5.freemalaysiatoday.com/category/nation/2017/06/09/uphill-climb-for-proton-despite-geelys-entry
14. WapCar – Proton 2025 sales. https://www.wapcar.my/news/proton-records-157976-vehicle-sales-in-2025-as-exports-reach-highest-level-since-2012-749204
15. Zigwheels – Proton 9M 2026 sales. https://www.zigwheels.my/car-news/proton-surpasses-160k-sales-in-9-months-two-new-models-still-to-come-this-year
16. Business Today – Malaysia auto competition 2026. https://www.businesstoday.com.my/2026/07/12/intense-competition-to-shape-malaysias-automotive-sector/
17. Carlist – Proton EV plant Tanjung Malim. https://www.carlist.my/news/proton-begins-construction-of-malaysia-s-first-dedicated-ev-plant-in-tanjong-malim-141304/141304/amp/
18. WapCar – AHTV and Geely brands. https://www.wapcar.my/news/chinese-state-media-confirms-drbhicoms-ahtv-to-ckd-more-geely-auto-group-brand-cars-79761
19. Malay Mail – DRB-HICOM returns to profit. https://www.malaymail.com/news/money/2022/11/24/drb-hicom-returns-to-black-chalks-up-rm14395m-net-profit-in-q3fy2022/41751